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Experiment Title : Demonstration of Modem Hardware and Signal Flow

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Experiment No. 04: Demonstration of Modem Hardware and Signal Flow

Aim

To identify the hardware interfaces of a Modem and demonstrate the flow between a PC and an ISP Server using Cisco Packet Tracer.

Theory.

1. Modem

A **modem (Modulator–Demodulator)** is an electronic communication device used to connect a computer or network to the internet through communication lines such as telephone lines, cable, or fiber.

The main function of a modem is to convert signals between two forms:

- **Digital signals** (used by computers)
- **Analog signals** (used in transmission over communication media)

A modem performs both **modulation** and **demodulation**, allowing data to be transmitted and received efficiently.

Functions of a Modem:

- Converts digital data into analog signals for transmission
- Converts analog signals into digital data for reception
- Establishes communication between user and internet service provider (ISP)

2. Working Principle of a Modem

The working principle of a modem is based on two key processes:

(1) Modulation

- The computer generates data in the form of **digital signals (binary: 0s and 1s)**
- The modem converts these digital signals into **analog signals**
- These analog signals are transmitted through communication channels like telephone lines or cable

(2) Demodulation

- The receiving modem gets **analog signals** from the transmission medium
- It converts these analog signals back into **digital form**
- The computer then processes the digital data

Working Flow:

Computer → Modem (Modulation) → Transmission Medium → Modem (Demodulation) → Computer

3. IP Address

An **IP Address (Internet Protocol Address)** is a unique numerical label assigned to each device connected to a computer network that uses the Internet Protocol for communication.

It serves two main purposes:

- **Identification of a device**
- **Location addressing (to send and receive data)**

Types of IP Addresses:

1. IPv4 (Internet Protocol Version 4):

- 32-bit address

- Written in dotted decimal form
- Example: 192.168.1.1

2. IPv6 (Internet Protocol Version 6):

- 128-bit address
- Written in hexadecimal form
- Example: 2001:0db8:85a3::8a2e:0370:7334

Importance:

- Helps devices communicate over the internet
- Ensures correct delivery of data packets

4. ISP (Internet Service Provider)

An **Internet Service Provider (ISP)** is an organization or company that provides internet access and related services to users.

ISPs act as a bridge between the user and the internet.

Functions of ISP:

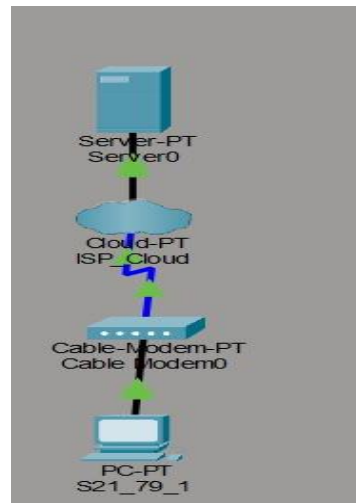
- Provides internet connectivity
- Assigns IP addresses to devices
- Maintains network infrastructure
- Offers services like email, web hosting, and broadband

Examples of ISPs:

- Jio

- Airtel
- BSNL
- Vodafone Idea

Block Diagram



Hardware Components Required

- **PC-PT:** The user's computer (Digital Source).
- **Cable Modem-PT:** The signal converter (Modulator-Demodulator).
- **Cloud-PT (ISP):** The Service Provider infrastructure.
- **Server-PT:** The remote destination (e.g., a Web Server).

Procedure

Step 1: Physical Topology Design

In this stage, we select the hardware from the device box and place them on the logical workspace.

- **Add End Devices:** Go to the bottom-left "End Devices" category. Drag and drop one PC (PC-PT) and one Server (Server-PT) onto the workspace. Rename the PC by your batch and roll number (e.g. S11_01).
- **Add the ISP Infrastructure:** From the "WAN Emulation" category, drag and drop the Cloud-PT between the PC and the Server. Rename this as "ISP_Cloud".
- **Add the Modem:** Go to "Network Devices" -> "WAN Emulation". Drag and drop the Cable Modem-PT between the PC and the Cloud.
- **Hardware Identification:** Click on the Modem and observe the physical rear view. Note the difference between the small RJ-45 Ethernet port and the threaded Coaxial port.



Step 2: Physical Cabling (Media Selection)

Selecting the correct transmission media is crucial for simulating real-world telecommunication links.

- **PC to Modem:** Select the Copper Straight-Through cable (solid black line). Connect the FastEthernet 0 port of the PC to Port 1 (ethernet port) of the Cable Modem.
- **Modem to ISP:** Select the Coaxial cable (solid blue line). Connect Port 0 (co-axial port) of the Cable Modem to the Coaxial 7 port of the ISP_cloud.
- **ISP to Server:** Select the Copper Straight-Through cable. Connect the Ethernet 6 port of the ISP_Cloud to the FastEthernet 0 port of the Server.

Step 3: ISP Cloud Logic Configuration

The Cloud must be told how to "route" the signal from the analog coaxial side to the digital server side.

- Click on the ISP_Cloud and go to the Config tab.

- In the left-hand menu, look under the INTERFACE heading and click on Ethernet6. Ensure "Cable" is selected under the Provider Network.
- In the left-hand menu, look under the CONNECTIONS heading and click on Cable.
- In the drop-down menus, select Coaxial 7 for the first field and Ethernet 6 for the second field.
- Click the Add button. You should now see Coaxial7 <==> Ethernet6 appearing in the mapping table.

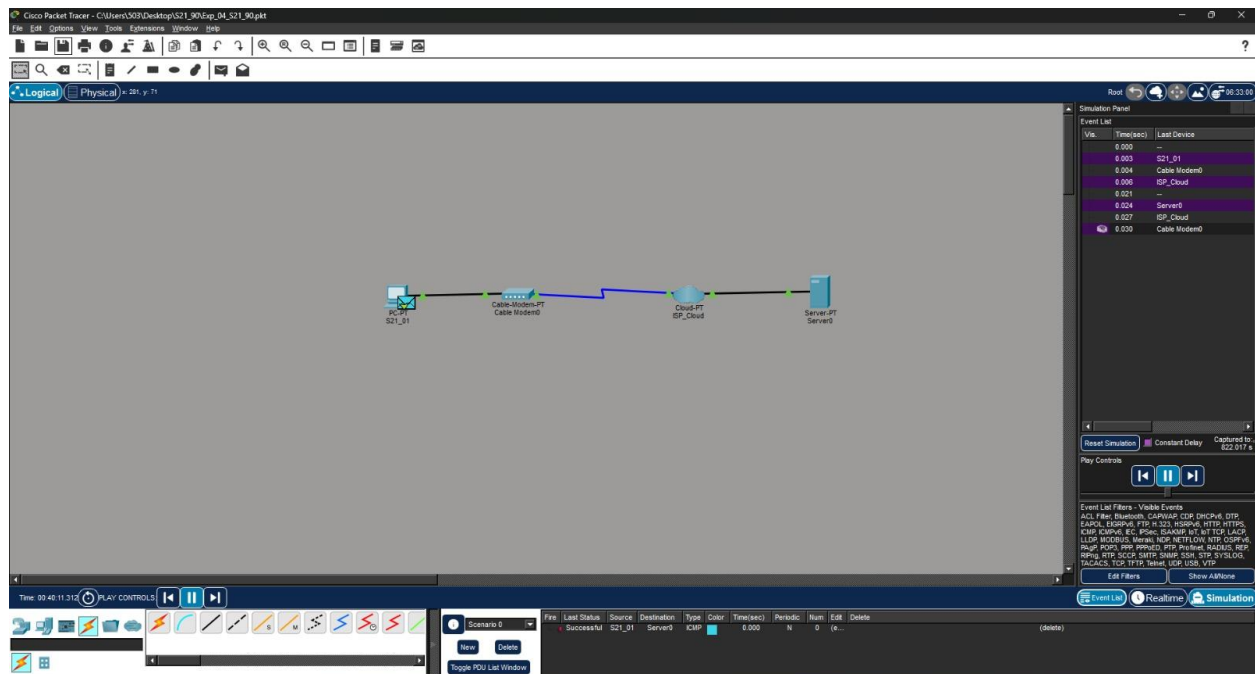
Step 4: IP Addressing (Logical Setup)

Both the source (PC) and destination (Server) must belong to the same logical network to communicate without a router.

- **Configure PC:** Click PC -> Desktop -> IP Configuration. Set the Static IP address to 192.168.1.1 and Subnet Mask to 255.255.255.0.
- **Configure Server:** Click Server -> Desktop -> IP Configuration. Set the IP address to 192.168.1.2 and Subnet Mask to 255.255.255.0.

Step 5: Testing and Packet Observation

- **Real-time Testing:**
 - Open the Command Prompt on the PC. Type ping 192.168.1.2. If you see "Reply from...", the modem is correctly modulating and demodulating the signal.
- **Simulation Mode:** Click the Simulation tab (bottom-right).
 - Click Edit Filters and select only ICMP.
 - Click the Add Simple PDU (Envelope icon), click on the PC, then click on the Server.
 - Click the Play/Forward button to observe the packet travel from the digital Ethernet link to the analog Coaxial link.



Observations

- **Medium Change:** It is observed that the data travels as a digital signal until it reaches the Modem.
- **Signal Conversion:** Upon hitting the Modem, the data is converted to travel over the **Coaxial medium** (simulating analog transmission).
- **Successful Loop:** The Server successfully receives the modulated signal, ISP cloud demaps the return packet back to the specific Coaxial line, allowing the PC to receive a successful reply.

Conclusion

The experiment successfully demonstrates that a Modem is an essential interface for internet connectivity. It proves that digital end-devices cannot communicate directly over analog WAN infrastructure without the modulation/demodulation hardware.